

When does cortical coherence carry shared content?

Compatibility and self-reconstruction as testable conditions

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Abstract

Synchrony accounts treat collective phase order as the mechanism that integrates distributed cortical activity. We show formally that shared content and a self-model each need a condition beyond phase order, and state it. Three conditions are separated: coherence (collective phase order), compatibility (agreement of independently decoded local descriptions where regions overlap) and self-reconstruction (accurate recovery of the global state from an internal encoding). Compatible local descriptions determine a unique global state. Coherence bounds disagreement between overlapping descriptions only through shared encoders of bounded sensitivity, and the bound grows as the square root of the population read: on one simulated cortical sheet it constrains at most 10 sites, chaining through overlapping patches extends it to 0.312 mm, and across distant regions the link from coherence to content is an assumption. A contracting self-map fixes one state but reconstructs accurately only states within a diameter set by its error and contraction factor. Lean 4 verifies these implications; one

model satisfies all connecting hypotheses at once, showing consistency, not cortical realization. The conditions yield a disconfirmable comparison: at matched phase coherence and single-region decoding accuracy, decoders fitted independently on two regions should agree more, beyond their separate accuracies, on perceived than on unperceived stimuli classified by a no-report marker. Choosing the overlapping regions by what each decodes makes the unity correlate operational relative to a pre-registered decoder and criterion. An exploratory EEG analysis is a measurement check only. Unity and a minimal self remain interpretations that need an independent experiential endpoint.

1 Introduction

Endogenous extracellular fields can affect spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015) and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction from an internal encoding). The mathematical states are spatial measures and fields, compared with perceptual content through independently fitted decoders (Section 10.2).

16 Binding-by-synchrony and communication-through-coherence accounts
 17 treat phase relations as the medium of integration or of selective routing
 18 (Singer & Gray, 1995; Fries, 2015), and field theories place integration in the
 19 endogenous electromagnetic field (McFadden, 2020; Pockett, 2012). Both
 20 leave open what fact about synchronized activity makes it one content rather
 21 than coordinated activity carrying several. The contribution here is to sep-
 22 arate the three requirements and prove which follow from which; each result
 23 names a measurement that phase data alone cannot supply.

24 The objection that synchrony does not fix content is older than this work.
 25 Critiques of the temporal binding hypothesis argued that synchrony, where
 26 it occurs, cannot say which features belong together or what they jointly
 27 represent: that grouping has to be computed before synchrony could sig-
 28 nal it (Shadlen & Movshon, 1999). The formal results here are elementary:
 29 uniqueness of gluing, Banach’s fixed-point theorem, the triangle inequality
 30 and a pigeonhole count. What formalization adds to the informal argument
 31 is quantitative and bookkeeping content. It gives the rate at which a co-
 32 herence bound on content weakens with population size, the population at
 33 which it stops constraining anything and how far chaining through overlap-
 34 ping patches extends it, the diameter within which one contracting map can
 35 reconstruct, and the number of codes a separated family of states requires.
 36 It also gives a complete inventory of the premises a synchrony-to-content
 37 argument consumes, each of which names a measurement, together with one
 38 model in which all of them hold at once.

39 The familiar noisy mean-field oscillator supplies a stationary coherence
 40 threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991); we add a minimum
 41 cost for changing coherence quickly. A Lean 4 development checks the im-
 42 plications, and a joint witness shows that the eight connecting hypotheses of
 43 the formal composition are jointly satisfiable.

44 The empirical question is which measurements could support or discon-
 45 firm the proposed connections. A phase resultant can miss a locally ordered
 46 winding field; stationary phase distributions can leave the coupling mecha-
 47 nism unidentified; and coherent adaptation can lose alignment with a desig-
 48 nated content template. We therefore distinguish stationary branches from
 49 finite observed onset, and compare prespecified field and connectivity onset
 50 models on held-out emergence recordings. The comparison tests a specified
 51 spatial prediction, not the presence of experience. Section 10.2 states the
 52 comparison that bears on conscious content directly.

53 The article makes six claims:

- 54 1. Compatible local descriptions determine a unique global state, while
 55 equal phases coexist with incompatible descriptions (Section 4).
- 56 2. Coherence bounds disagreement between overlapping descriptions
 57 through shared encoders of bounded sensitivity, by $L\sqrt{2N(1-r^2)}$
 58 on the population read; on a simulated cortical sheet this constrains
 59 at most 10 sites, and chaining through overlapping patches extends it
 60 to 0.312 mm (Section 4.2).

- 61 3. A contracting self-map fixes one state and reconstructs within error ε
62 only a family of diameter at most $2\varepsilon/(1 - \Lambda)$; reconstructing M states
63 pairwise more than 2ε apart takes M distinguishable codes (Section 5).
- 64 4. Changing coherence in time τ has a minimum current cost that grows
65 as $1/\tau$ (Section 3.2).
- 66 5. Stationary phase statistics leave the coupling mechanism unidentified,
67 finite ramps shift the observed onset, and a winding field keeps local
68 order at global $r = 0$ (Section 8).
- 69 6. The link to content predicts that, at matched phase coherence and
70 single-region accuracy, decoders fitted independently on two regions
71 agree more, beyond their separate accuracies, on perceived than on
72 unperceived stimuli; equal excess counts against the link to content
73 (Section 10.2).

74 Sections 2–5 develop the three requirements; Section 6 applies their mea-
75 surement obligations to a digital candidate. Section 7 states what the formal
76 composition establishes, and Sections 8–9 examine observables and the corti-
77 cal test, and Section 11 states the scope of every result in one place. Table 1
78 lists the connecting hypotheses the article argues from; Supplemental Mate-
79 rial gives the full chain, derivations and controls. For a theoretical article the
80 Methods are the definitions, proofs and simulations. Each result is stated in
81 the section that uses it, Supplemental Material Table S1 maps it to its Lean
82 declaration, and every simulation’s model, parameters and controls are given

83 in Supplemental Material. The Results are the theorems of Sections 3–5 and
the numerical controls of Section 8.

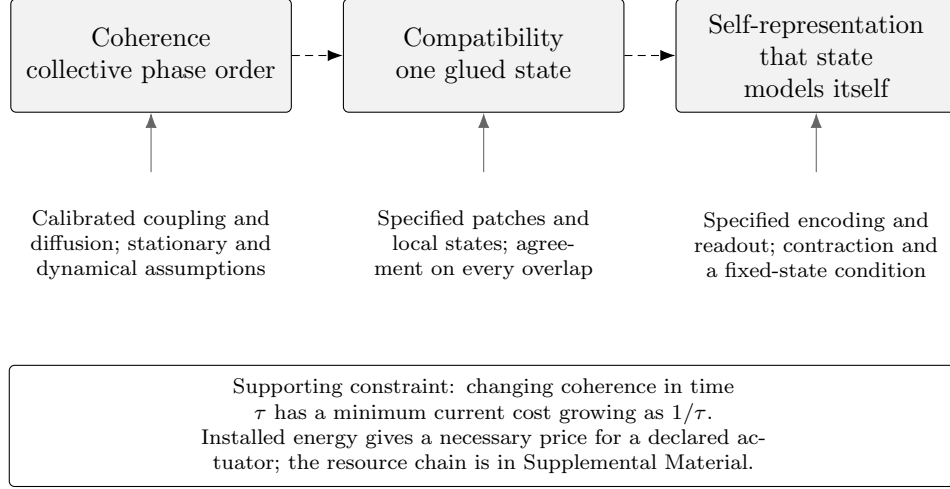


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections, each of which needs the inputs of the condition it points to in addition to phase coherence. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the connecting hypotheses of the formal composition that the article argues from.

84

85 2 Limits of phase order

86 Several limits hold without the cortical hypothesis. Equal phases can coexist
87 with incompatible local descriptions; phase order constrains shared content
88 only through encoders related on overlaps. A winding field can have zero
89 global resultant while remaining locally ordered. A constant readout can have
90 a unique fixed point without reconstructing its input. A region encodes only

the states it distinguishes, and by a declared deadline a site can report only changes inside its causal past. These limits reflect familiar phase oscillator and information constraints (Acebrón et al., 2005; Ott & Antonsen, 2008; Dörfler & Bullo, 2014; Breakspear, 2017). The sections below state each one beside the condition or measurement it motivates.

3 Coherence: physical coupling and its cost

3.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct parameters. The stationary calculation below uses identical frequencies and this scalar coupling. Applying it to a spatial kernel requires an aggregation rule and a check of the resulting approximation.

For identical frequencies in their rotating frame, the positive normalized

stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

where I_n are modified Bessel functions and ψ is the collective phase. The formal derivation uses the absence of stationary current for a periodic gradient drift to obtain this density, then verifies normalization and self-consistency. At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz & Mirollo, 1991). Above threshold, exactly one positive solution coexists with zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

Figure 2 shows this stationary branch.

Linearization about the uniform density gives rate $K/2 - D$ for the first harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists exactly above threshold. About the coherent density, the quadratic form is strictly dissipative transverse to collective rotation, whose rate is zero. These linear estimates concern the limiting equation. Relating them to finite stochastic trajectories requires a dynamical mean-field limit and branch selection; exchange symmetry alone does not establish the needed independence (Sznitman, 1991).

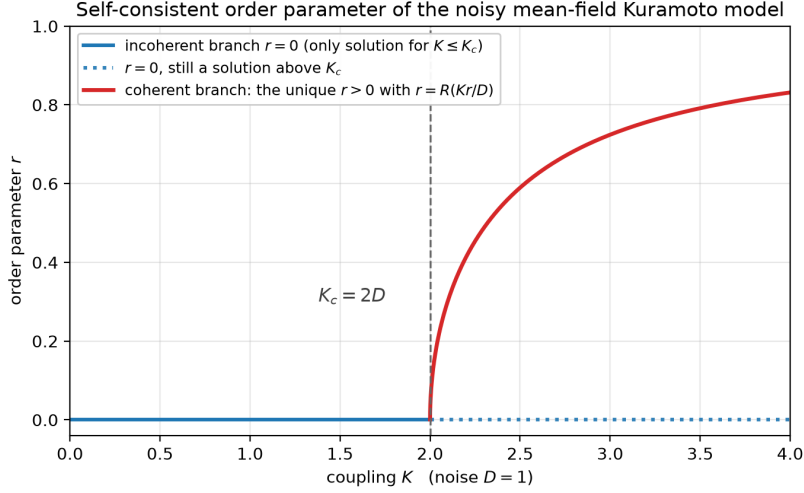


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

123 3.2 A speed limit on changes of coherence

124 One thermodynamic statement bears on coherence directly. For a smooth
 125 positive periodic phase density with probability current J and diffusion D ,
 126 the current functional $\sigma_J = \int J^2 / (D\rho) d\theta$ satisfies

$$\int_0^\tau \sigma_J dt \geq \frac{[\arcsin r(\tau) - \arcsin r(0)]^2}{D\tau} \quad (4)$$

127 while $0 < r < 1$. A change of coherence completed in time τ therefore has a
 128 minimum current cost that grows as $1/\tau$, whereas holding a stationary order
 129 has no positive floor. Read the other way, a current budget C sets a minimum

130 duration $\tau \geq [\Delta \arcsin r]^2 / (DC)$ for a given change. This bears on ignition-
 131 like transitions, in which large-scale coherence rises abruptly (Dehaene et al.,
 132 2014), and on recovery latency after anaesthesia: each has a floor set by the
 133 current available to drive it. The bound is a heat or power statement only
 134 when J is identified with a physical flux, a reservoir at a stated temperature
 135 is specified and local detailed balance holds for the steps concerned; only
 136 then does σ_J convert to entropy production (Landauer, 1961; Norton, 2005;
 137 Shenker, 2000). Installed energy supplies a separate, necessary price: for
 138 a declared actuator whose response modes carry installation prices, mean
 139 coupling is at most κU_{inst} for a mode-data factor κ , so a coherent stationary
 140 branch requires installed energy $U_{\text{inst}} > 2D/\kappa$. Supplemental Material gives
 141 both derivations, the register and feedback budgets, and the sensitivity of
 142 the metabolic comparison to its assumptions.

143 **4 Compatibility: from coordinated activity** 144 **to a global description**

145 **4.1 Why common timing is insufficient**

146 The formal setting is a sheaf: each patch of the substrate carries local de-
 147 scriptions, and restriction maps cut a description on a patch down to any
 148 smaller patch. A *section* is a description on one patch, and a *global section* is
 149 a description on the whole substrate. For overlapping patches U and V , com-

150 patibility means that restricting either local description to $U \cap V$ gives the
151 same object. Descriptions can agree on shared quantities while contributing
152 different information elsewhere. Compatible sections of the specified sheaf
153 have a unique global extension. On a finite cover, local finite spatial mea-
154 sures agreeing as measures on every overlap also have a unique finite-measure
155 extension whose sheaf image is the glued section. No mass-one normalization
156 is imposed.

157 In the cup example, visual and reaching descriptions must express a
158 shared quantity, such as cup location, in common coordinates. Restriction
159 retains that quantity while dropping details particular to each description.
160 Agreement permits surface appearance and hand configuration to contribute
161 to one situation. The mathematical counterexample assigns the same phase
162 to every patch while two local mass profiles disagree on a shared site. Those
163 profiles have no common extension. Because they are chosen independently
164 of phase, synchrony places no constraint on their disagreement.

165 This distinction limits the biological interpretation. Receptive territories
166 and content domains must be related to spatial patches, and the descriptions
167 must support the required extension. Matching overlap marginals of de-
168 coded probability laws need not determine a unique global joint distribution
169 (Abramsky & Brandenburger, 2011).

170 The cover is itself a choice, and the construction constrains it in one way
171 only: synchronized patches agree where they overlap. Where they do not
172 overlap there is nothing to compare, so a cover by pairwise disjoint patches

173 satisfies that obligation for *every* assignment of local descriptions and phases
174 and the gluing theorem still returns a unique global section, which is the
175 tuple it was handed. Membership of the class then reports nothing about the
176 system beyond the patches covering it. What a gluing claim asks of a cover
177 is therefore a non-degeneracy the theorem does not assume: that distinct
178 patches meet, and that what they share is a quantity fixed independently of
179 the agreement it is being used to establish.

180 The cup example exposes a second gap, between two kinds of overlap.
181 The sheaf is built over a spatial substrate, so its patches overlap where they
182 share sites. Visual and motor territories that agree on cup location overlap
183 in what they describe, not in where they lie, and the construction supplies
184 no relation between the two. One route recasts the construction over content
185 variables: each region becomes a partial decoder whose domain is the set of
186 variables it estimates, and two regions overlap exactly where they estimate
187 a common variable. Gluing and approximate selection hold on such a cover,
188 and there compatibility is a consequence: regions decoding a variable within a
189 tolerance agree on it within twice that tolerance, so only agreement beyond
190 fidelity is evidence. The phase-to-content bounds concern spatial patches
191 and do not transfer. One principle selects such a cover without presupposing
192 the agreement it is used to test. Assign each region the variables that can
193 be decoded from its activity above a declared criterion, on data held out
194 from the agreement test, and let two regions overlap on exactly the variables
195 decodable from both. Decodability is fixed by what each region carries before

any comparison is made, so the cover cannot be chosen to make agreement come out. It is non-degenerate exactly when its nerve is connected, meaning that every region is linked to every other through a chain of shared decodable variables, and recordings can fail that condition. The principle has three inputs, the candidate variables, the criterion and the decoder class, and each has a rule that does not consult agreement.

The candidate variables are those the experiment manipulates and the task makes behaviourally relevant, fixed before recording. Data-driven alternatives, such as directions of maximal shared variance between two regions, select for the agreement under test and are excluded. The criterion becomes a statistical one: a region encodes a variable when its cross-validated decoding exceeds a permutation null at a pre-registered level, and results are reported across a declared range of levels. The level need not be exact, because two accurate decoders agree automatically (Eq. (13)); the statistic of Section 10.2 is agreement in excess of what each region’s accuracy implies, which remains valid at any admitted level. The decoder class is bounded on both sides. Unrestricted nonlinear decoders recover a variable from any region carrying information about it, so nearly every pair of regions overlaps and the cover again constrains nothing. The positive bound of Section 4.2 requires encoders with a declared Lipschitz constant L , which enters δ_r directly, so the class should consist of L -Lipschitz maps; linear readout, the usual operationalization of information explicitly available to downstream neurons (DiCarlo & Cox, 2007; Kriegeskorte & Diedrichsen, 2019), is one such class. Within the

219 admitted family, the decoder and the distance in which content discrepancy is
 220 measured are chosen by held-out decoding accuracy and never by agreement.
 221 Neither is neutral: which similarity measure best describes neural activity
 222 differs across tasks (Bobadilla-Suarez et al., 2020). What remains is that the
 223 cover is the experimenter’s. Whether cortex realizes unity over the chosen
 224 variables is the empirical claim under test, and no choice of decoder settles
 225 it. The correlate is therefore operational relative to these pre-registered in-
 226 puts, and carrying the formal results to such a cover is the recasting sketched
 227 above.

228 **4.2 When coherence does constrain content**

229 A declared relation between phase and content gives a positive result. Let
 230 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 231 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 232 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance,
 233 meaning that an encoder’s output moves by at most L times the chord
 234 between two input phases, and require $e_{i,x} = e_{j,x}$ wherever patches i and
 235 j overlap. Encoders may vary across sites; their equality on overlaps is a
 236 supplied relation between descriptions of the shared quantity. Write $z_k = e^{i\theta_k}$
 237 and $m = N^{-1} \sum_k z_k$, so $|m| = r$. The phasors’ squared deviations from their
 238 mean sum to $\sum_k |z_k - m|^2 = N(1 - r^2)$, and $|z_i - z_j| \leq |z_i - m| + |z_j - m|$
 239 gives $|z_i - z_j|^2 \leq 2N(1 - r^2)$ for each pair. Consequently every overlap

240 discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1-r^2)}. \quad (5)$$

241 Here discrepancy is the norm of the difference, and r and N are read on
242 whichever population the bound is applied to. That choice is what decides
243 whether the estimate says anything. A patch carries an order parameter
244 of its own, and reading the bound there carries that patch's resultant and
245 that patch's site count, so a large population observed through small patches
246 is constrained by the small count. Read instead on the whole array, the
247 factor $\sqrt{N}$ makes the estimate weak, because it draws a pointwise guarantee
248 from an average over N sites. The order cannot be improved in general:
249 one oscillator in antiphase among N leaves $N(1-r^2)$ near 4. At $r=1$ the
250 observations agree exactly at either scale. A shared scalar encoder is a special
251 case. Dropping the overlap relation permits disagreement even at perfect
252 locking. A common nonexpansive linear projection preserves agreement.

253 What Eq. (5) bounds is L times a chord between phasors, and a chord
254 never exceeds 2, so the estimate constrains anything only where $\sqrt{2N(1-r^2)}$
255 falls below 2, whatever the value of L . On the simulated sheet of Section 8,
256 at its patch-local order for a coupling decay length of 0.1 mm, the bound is
257 informative for at most 10 sites; this count and the chain's reach below are
258 values for that sheet and move with the decay length and patch order. Three
259 conditional results decline the step that puts $\sqrt{N}$ there. A pair count bounds

260 the fraction of disagreeing pairs at any population size but names no pair.
 261 A chain across the sites that overlapping patches share compares distant
 262 territories through the patches between them, with error growing linearly in
 263 hops rather than with population: on the same sheet it stays informative
 264 for 19 hops, 0.312 mm. A spectral gap removes the population count at a
 265 frequency-locked configuration, given a declared algebraic connectivity of the
 266 coupling graph. Territories whose agreement unity concerns lie centimetres
 267 apart, beyond the chain's reach, and across them the coherence-to-content
 268 link is the bridge assumption E78 (Table 1). Supplemental Material gives
 269 the three derivations and the sheet's numbers.

270 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 271 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 272 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 273 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (6)$$

274 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 275 duce an initial mismatch. Approximate observations leave a nondecaying
 276 term in the error bound. Taking a uniform bound on Eq. (5) connects this
 277 update to the phase process.

278 4.3 What approximate agreement determines

279 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
280 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
281 Weights that vanish outside their patches and sum to one at each site select
282 a global profile by weighted averaging. The selection lies within ε of each
283 patch there, and any two such selections differ by at most ε , independently
284 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
285 $L\varepsilon$ across selections.

286 At zero tolerance, the selection is the unique exact extension. At every
287 positive tolerance there are families with no exact extension; selection then
288 gives an approximation whose dependence on weights is bounded. Supple-
289 mental Material gives a three-site witness attaining the bound. The mass-
290 profile construction uses the scalar case of the content dynamics and addi-
291 tionally requires nonnegativity of the initial profiles and observations.

292 These constructions concern declared objects: a specified cover, contents
293 valued in a normed space, and encoders related on overlaps. Section 8 sets
294 out what a measurement of their neural counterparts would have to establish.

295 5 Self-representation: a state that models it- 296 self

297 5.1 The reflexive construction

298 Compatibility gives a global state. Reflexivity relates it to an encoding within
299 that state. Equip global sections with a complete metric. *Restriction reso-*
300 *nance* is the additional condition that the encoding is the state’s own restric-
301 tion to a region $\mathcal{A}$, so that the region encodes the state by being part of it.
302 Under that condition the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (7)$$

303 The region and readout are fixed across the state space. If F contracts and
304 fixes the section s_* formed from the local descriptions, Banach’s theorem
305 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
306 (Table 1 gives the full composition and its edges) supplies the contraction
307 and $F(s_*) = s_*$; restriction resonance is separate. Supercritical coupling
308 makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ less than one for a
309 relaxation time $\tau > 0$; Section 11 states the standing of that factor. A fixed
310 contraction can reconstruct accurately only a restricted family of states: if
311 its error is at most ε throughout that family, any two members are at most
312 $2\varepsilon/(1 - \Lambda)$ apart.

313 In the cup example, the global state includes the represented scene and the

314 perceiver's relation to it. A local encoding must retain enough information
 315 for the readout to recover that modelled state; cup position alone would be
 316 insufficient if hand configuration and viewpoint could not be reconstructed.

317 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-
 318 tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (8)$$

319 At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for
 320 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 321 weakens.

322 5.2 Reconstruction across a family of states

323 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 324 encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that
 325 family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (9)$$

326 Thus M relevant states pairwise more than 2ε apart require at least M
 327 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 328 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 329 existing self-prediction map.

330 This is an additional accuracy criterion on a family, with no requirement
 331 for multiple exact fixed points. For the same fixed contracting map used by

332 E89, however, the diameter bound above is an additional restriction: at zero
 333 error the family contains at most one metric-distinct state. A broad family
 334 needs an input-conditioned map, one contraction F_u per input u . If F is
 335 L -Lipschitz in u , members with inputs u, v lie within $(2\varepsilon + L d(u, v))/(1 - \Lambda)$
 336 and the fixed points within $L d(u, v)/(1 - \Lambda)$, so a self-state can follow the
 337 scene. The code criterion then binds within each input: a readout storing
 338 one state per input reconstructs without encoding anything. Under the fixed
 339 map of E89 the minimal-self correlate is a condition on one self-state. A blind
 340 encoder or constant readout fails the criterion for separated states despite
 341 having a unique fixed point.

342 Where the encoding is read as the state's own restriction to a region, the
 343 criterion becomes a condition on the region. Restriction resonance requires
 344 the encoder and that restriction to identify the same pairs of states, so a
 345 boundary reconstructing two relevant states more than 2ε apart while its
 346 region reads them identically is not resonant. Read forwards, a resonant
 347 boundary that reconstructs a pairwise separated family restricts injectively
 348 on it. Which states a region reads identically is fixed by where the region
 349 sits, before an encoding is chosen, so the resolution the family demands is
 350 the region's and no readout supplies it. The implication runs one way: a
 351 region that does resolve the family is not thereby shown to carry a resonant
 352 encoding.

353 Resonance and resolution are jointly satisfiable without a local fold. Take
 354 the region to be the whole substrate and use the identity readout: restric-

tion is the identity, so this boundary is resonant and reconstructs every de-
 clared family exactly. Its map has Lipschitz factor one and does not satisfy
 E89’s strict contraction premise on a nontrivial metric space. A properness
 condition on the region is therefore a separate requirement, and the fixed-
 contraction diameter bound remains in force when E89 is also required. With
 an L -Lipschitz readout, a family reconstructed within ε whose members lie
 pairwise more than $d_{\min}$ apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$,
 so the family is no larger than the δ -packing number of what the region car-
 ries. At the whole substrate it says nothing, because the region’s range is
 everything. The gain L is a declared input, and the bound weakens without
 limit as the gain grows.

5.3 Agreement after a change takes communication

Snapshot compatibility does not specify how quickly descriptions respond
 when a shared quantity changes. In a synchronous network whose updates
 read only the site’s own state and declared neighbours’ messages, two initial
 configurations agreeing on a site’s causal past through T rounds produce the
 same state there at round T . The deadline consequence is the one Section 2
 states, and it follows from the reconstruction bound in Section 5.2: a report
 from that site is a readout of a code that the two interventions leave identical.

A three-site delay line witnesses failure after one round and exact report-
 ing after two; cutting the path prevents delivery at every round. A report
 may still happen to be correct about one fixed world. The bound concerns

377 responsiveness across interventions.

378 Throughout this section *prediction* names a reconstruction map, and the
379 relevant family of states, the metric, the tolerance, the communication graph
380 and the deadline are all declared inputs, as are the represented variables and
381 the fidelity counted as accurate. Reading the map as temporal neural pre-
382 diction would require dynamics; reading the resulting code count as physical
383 memory capacity would require a hardware model, which a state-dependent
384 readout would have to extend.

385 **6 What the conditions ask of a digital candi-** 386 **date**

387 A claim that a digital system, such as a large language model (LLM) run-
388 ning on graphics processing units (GPUs), has unified content or represents
389 itself commits its maker to the measurements this account asks of cortex.
390 The maker must identify the shared quantities the system’s parts describe,
391 show that decoders fitted independently on those parts agree about them,
392 and show that a declared local code separates the relevant family of states
393 and survives intervention. The indicator-properties approach assesses arti-
394 ficial systems against computational properties drawn from theories of con-
395 sciousness (Butlin et al., 2023); the conditions here are fewer and each is
396 a measurement rather than an architectural property. A digital system is
397 also where they can be measured today, with full access to its state and

398 exact interventions on it: fitting decoders for one shared quantity on two
399 separate layers, heads or token positions, and testing their agreement and
400 whether reconstruction survives intervention, would carry out the proposed
401 measurement directly.

402 One constraint follows from the architecture alone. Within one forward
403 pass of a causally masked model, position t reads only its causal past, so
404 an intervention outside that past cannot be reconstructed at t by that step.
405 This is the deadline result of Section 5 applied to the mask, and it bounds
406 which variables a position can encode about the rest of the sequence within
407 one step. A second constraint needs a premise the architecture does not
408 supply: a linear encoder and readout factoring through d coordinates pay
409 reconstruction error at least the discarded spectral tail of the coupling oper-
410 ator, but query/key dimension does not supply such a bottleneck for softmax
411 attention. These results rank no architecture family, imply no general infe-
412 riority of GPU hardware and settle no question of machine experience. A
413 companion paper (Bobadilla-Suarez, 2026) adds a condition on how agree-
414 ment is produced: it proposes that unity requires agreement enforced by a
415 coupling through which the graded physical state of each region pulls the
416 regions it overlaps toward agreement, and proves that content read through
417 noise margins, as content read from bits is, never meets that condition. Un-
418 der that premise the measurements proposed here are necessary for unity in
419 any substrate, and a digital system that passes them computes its agreement
420 rather than enforcing it. Supplemental Material gives the bottleneck deriva-

421 tion, a synthetic two-view task with its controls, the information ceiling of
422 token outputs and the hardware scope.

423 **7 The formal composition**

424 The Lean development composes the three requirements into one conditional
425 theorem. Its passive and feedback branches each accept eight connecting hy-
426 potheses, running from finite capacity through a register, dissipation and a
427 continuum coupling to the field, coherence, the cover and the self-prediction
428 fixed point. Each edge is a hypothesis of the composition, and its inferen-
429 tial content is its last step, where contraction and a fixed glued section give
430 uniqueness. Table 1 lists the four edges the article argues from; Supplemen-
431 tal Material gives the full chain, its thermodynamic edges and the register
432 ledger of the feedback branch. Both branches also take a cover carrying ex-
433 act overlap compatibility and a reflexive boundary specifying encoding and
434 readout, and conclude that the glued section is the unique fixed point of self-
435 prediction. Composing the edges establishes that the premises are consistent,
436 named in one place and machine-checked together, and joint witnesses satisfy
437 every edge of each branch at once. The development declares no additional
438 axioms beyond its libraries' foundational axioms.

Table 1: The four connecting hypotheses the article argues from, shared by both branches of the formal composition. Supplemental Material gives the full eight-edge chain and, in Table S1, maps the node-level results.

Connection	Status	Required assumption
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover’s nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve’s diameter; where the configuration is frequency-locked it carries through the coupling graph’s algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.

439 8 What the observations establish

440 The composition holds under its premises. Whether observation can estab-
441 lish those premises is a separate question, and the three requirements it
442 connects concern different objects: phase dynamics, local descriptions and
443 internal encodings. Their observables must retain these distinctions. The
444 following controls show which mechanisms remain indistinguishable, which
445 temporal approximations fail, which states one observable cannot separate,
446 and what independent content measurements must establish. Each is a finite
447 numerical regime on a declared model, with full parameters and controls in
448 Supplemental Material, and none of them calibrates a cortical quantity.

449 8.1 Stationary distributions leave the mechanism un- 450 determined

451 Equation (2) gives a concentration–coherence relation with no fitted shape
452 parameter. Estimate r from circular resultant length and a from density
453 shape: inverting R from the measured r would make agreement automatic.
454 Even independently estimated agreement identifies a statistical family shared
455 by different mechanisms.

456 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
457 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
458 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
459 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,

460 and diffusion D_0 have that same density. Their stationary distributions co-
 461 incide window by window despite different causes.

462 Resolved dynamics supply additional information. In the ideal model
 463 with continuous latent paths, quadratic variation accumulates at rate $2D$
 464 and conditional increment means constrain drift; a filtered observation of
 465 the phase needs its own observation model before either is read off. Separat-
 466 ing endogenous coupling from a matched common drive additionally requires
 467 a measured drive or controlled perturbation; identifying electromagnetic cou-
 468 pling requires evidence specific to that interaction. Supplemental Material
 469 gives the equivalence construction and its numerical check. Phase statistics
 470 pooled over time and over bipolar pairs cannot supply these estimates: on
 471 public sedation EEG (Bajwa et al., 2025), the pooled pipeline places a per-
 472 fectly phase-locked signal at $r = 0.0001$, so a spatial test needs an estimand
 473 that removes the common rotation and a montage whose effect on spatial
 474 phase is modelled; Supplemental Material reports this estimator exercise and
 475 the dependence-aware design.

476 **8.2 Finite ramps change the observable onset**

477 The square-root law in Eq. (3) describes stationary solutions, and a changing
 478 coupling needs time for the phase distribution to approach them. In a deter-
 479 ministic mean-field control crossing the threshold, none of the scanned rates
 480 meets the declared tracking tolerance at onset. In a finite-population ramp
 481 study ($N = 2000$, $D = 1$, 32 replicas, 8 speeds), faster ramps delay escape.

482 The fitted delay exponent 0.444 has a replica-bootstrap interval $[0.398, 0.496]$
 483 that excludes the ideal $1/2$, a deterministic reference shows that the escape
 484 criterion changes the fitted exponent, and no common asymptotic exponent
 485 is established. Delayed passage through a noisy bifurcation (Berglund &
 486 Gentz, 2002) makes the apparent exponent depend on ramp speed and fit-
 487 ting window. Supplemental Material gives the sweep and its figure, the
 488 fluctuating-coupling and population-size controls, and the signed-network
 489 crossing bracket.

490 **8.3 A winding state leaves the order parameter am-** 491 **biguous**

492 The global order parameter r discards spatial arrangement. On a periodic
 493 sheet, a phase field can wind once around the sheet and have $r = 0$ while
 494 neighbouring sites remain locked. At identical frequencies, every such wind-
 495 ing is stationary for an isotropic symmetric kernel: contributions at opposite
 496 separations cancel. It violates the quarter-turn initial-spread premise of the
 497 convergence result, which is why that result can still select the uniform state.

498 In the spatial control, the winding persists below the interpolated de-
 499 cay length 0.1871 mm and unwinds above it (Fig. 3). At 0.1 mm, uniform
 500 and winding starts under the same coupling, diffusion and kernel end at
 501 $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched fre-
 502 quency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm,

503 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen
 504 realizations. These are finite controls, not a cortical wave identification or a
 505 derived transition.

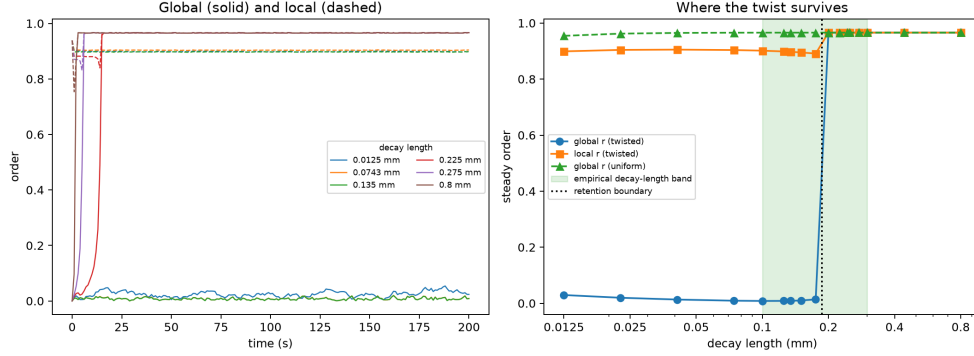


Figure 3: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked.

506 Linearization about a winding gives explicit mode rates, which concern
 507 linear stability rather than the measured retention boundary. An amplitude
 508 extension adds an existence condition under which a broad kernel can remove
 509 a winding that the phase-only equations retain as a stationary state.

510 Neither model turns these states into measured cortical spirals. The pe-
 511 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has
 512 no propagation speed, and the amplitude variable has no calibrated relation
 513 to a recorded envelope. Supplemental Material gives the spectral, amplitude

514 and defect results. Locally, Eq. (5) can still bound content disagreement on a
515 winding patch when shared encoders are supplied, although its global $r = 0$
516 form is uninformative. A nonzero ring winding number q also obstructs a
517 global real-valued phase lift; it has no identified content meaning here. Thus
518 an onset criterion based only on r can register a change of basin as well
519 as a change of coupling. The protocol of Section 9.3 needs a spatial phase
520 observable.

521 **8.4 Coherent adaptation can lose environmental align-** 522 **ment**

523 An adaptive coupling rule might supply representational structure, but its
524 objective must reward the relevant agreement. For symmetric coupling, pair-
525 wise sine terms cancel in the population mean drift. Minimizing squared
526 drift at that fixed mean therefore rewards equalizing drift across nodes. In
527 the frequency-cluster control, this favors coupling between unlike-frequency
528 clusters, whereas the diagnostic environmental template rewards coupling
529 within clusters. Supplemental Material proves descent of this objective for
530 a specified gradient flow at fixed phases and distinguishes it from entropy
531 production.

532 The finite control co-evolves noisy phases and symmetric, nonnegative
533 couplings at fixed total resource. Its template is a diagnostic covariance
534 of latent signals, never a training input. Across three illustrative seeds,

second-half order remains between 0.9167 and 0.9222, while the within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4). Random-update, permuted-gradient and frozen-kernel controls are reported in Supplemental Material.

The immediate post-update objective reduction disappears when averaged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

8.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on synthetic data separates these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial;

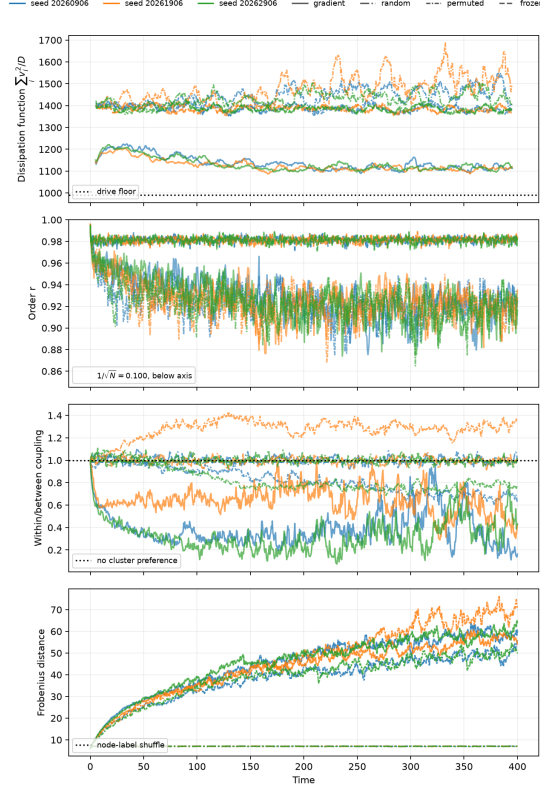


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

557 the compatible construction scores 0.476 on an interval containing one half.
558 Shared-prior shrinkage can nevertheless manufacture agreement, and regional
559 bias can manufacture disagreement. Per-patch decoding diagnostics detect
560 these failures; the compatibility statistic alone does not.

561 On constructed content, phase carries a share 1.000 of the content entropy
562 when content is built from it and 0.000 when it is not. A phase-derived statis-
563 tic reports compatibility on phase-locked patches whose local descriptions
564 disagree. The implemented content statistic detects their incompatibility.
565 These controls validate a measurement on known constructions; a neural
566 content model and decoder remain to be established.

567 Choosing the cover also chooses a candidate subject: trivial descriptions
568 can agree while carrying little content, and a partition can conceal disagree-
569 ment. Section 4 gives the decodability rule that fixes the cover without
570 consulting agreement.

571 **9 The cortical hypothesis and an awakening**

572 **test**

573 The cortical proposal identifies a physical source of coupling and asks whether
574 its changes help explain recovery. The tests in Section 8 determine what
575 observations must distinguish before a phase trace can support this identifi-
576 cation.

577 9.1 Physical identification and calibration

578 The cortical hypothesis identifies part of effective coupling with endogenous
579 extracellular electromagnetic interactions. Fields can modulate spike timing
580 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
581 propose coordination of distributed neural information (Pinotsis & Miller,
582 2022; McFadden, 2020). Phase organization under propofol motivates study-
583 ing this regime (Bastos et al., 2021).

584 The formal field predicate requires a jointly continuous kernel, a normal-
585 ized substrate, coupling equal to the kernel average, and no single site pro-
586 viding more than half that strength. Toy constructions satisfy it. A cortical
587 realization must specify the scale at which the kernel and its reduction ap-
588 ply: calcium imaging reports different state-dependent modularity patterns
589 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

590 Cortical columns are a serious candidate for the oscillator index. They are
591 delineated by response properties and anatomy rather than by the agreement
592 they would be used to establish (Mountcastle, 1997), and a phase can be
593 estimated from a column-sized territory at the electrode spacings used for
594 cortical wave measurements (Townsend et al., 2015). They are a poor content
595 cover: non-overlapping columns form a disjoint cover (Section 4), much of
596 association cortex has no accepted columnar parcellation, and the column’s
597 standing as a canonical unit is contested (Horton & Adams, 2005; Rakic,
598 2008). The content cover is the decodability cover of Section 4, and the
599 bridge assumption E78 records its identification with the oscillator index.

600 The spatial reduction is also testable. A row-sum aggregation at zero
601 diagonal tracks the uniform-coupling control over the tested spatial decay
602 range, while frequency heterogeneity displaces the operational threshold. The
603 fit window biases both estimates relative to $2D$; agreement with the control
604 is therefore the supported conclusion. Supplemental Material reports the
605 reduction, its numerical error and the distinct assumptions needed to identify
606 the kernel average with the scalar coupling of the installed-energy condition.

607 Converting a spike-time shift into coupling needs a rate: the dimensionless
608 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
609 frequency f becomes a coupling through a conversion rate γ ,

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (10)$$

610 This rate describes how induced timing shifts accumulate into drift.

611 **9.2 Extracellular geometry and recovery**

612 The proposed control variable is extracellular geometry. Extracellular vol-
613 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
614 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
615 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
616 at $0.3\text{--}0.6 \text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
617 an effective-medium description predicts that a larger extracellular fraction
618 raises conductivity and reduces field amplitude.

619 The coupling-gated hypothesis consequently predicts that extracellular
620 contraction on awakening raises E and, holding the other factors in Eq. (10)
621 fixed, raises effective coupling. This argument supplies a direction; the
622 geometry–coupling relation, its magnitude and its role in human recovery
623 require measurement.

624 For a smooth crossing at nonzero speed, evaluating the stationary branch
625 along the changing control gives a deterministic tracking approximation that
626 fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (11)$$

627 This is a deterministic tracking approximation from Eq. (3) at fixed D ; the
628 crossing controls in Section 8.2 show that it fails as an observed-onset ap-
629 proximation under the tested initialization. A separation of geometry and
630 phase time scales can support tracking away from threshold. The awakening
631 hypothesis needs a calibrated dynamical prediction for an operational onset
632 above the finite-size fluctuation floor. Crossing K_c alone does not supply a
633 recovery latency.

634 **9.3 A protocol for the awakening experiment**

635 Neural inertia describes a behavioural barrier and hysteresis between anaes-
636 thetic induction and emergence (Friedman et al., 2010); recovery can pass
637 through metastable activity states (Hudson et al., 2014; Proekt & Hudson,

2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field mechanism from a synaptic one. A possible discriminator is spatial onset relative to independently estimated local field coupling and network connectivity. The field-based onset map below is a separate model from the mean-field one, specified before the observations are inspected.

For that comparison, partition the recording into prespecified regions i and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements, and a connectivity score c_i from a separately estimated structural network. For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic approximation predicts

$$\hat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (12)$$

Predicted onset is the first crossing of a region-specific threshold fixed from pre-onset fluctuations. Fit the two scales on training sessions, then compare held-out onset times and site order under the same observation model. Larger field scores predict earlier onset only when baseline order and diffusion do not reverse the ranking. This declared approximation omits interregional coupling and saturation; a successful ranking would support its field covariate, not prove ephaptic causation. Supplemental Material specifies the

658 controls and failure conditions.

659 A within-subject protocol can manipulate emergence rate, with spatially
660 resolved recordings to compare the proposed field and connectivity models:

- 661 1. **Manipulate emergence rate.** Vary the propofol infusion taper
662 across sessions within the same subject, recording drug concentrations
663 or a validated pharmacokinetic proxy to quantify the manipulation.
- 664 2. **Measure behavioural recovery.** Record time-to-response to a stan-
665 dard stimulus as the independent behavioural endpoint, rather than
666 estimating ΔK directly.
- 667 3. **Resolve spatial onset.** Identify anatomical locations where coher-
668 ence first crosses the baseline-scaled threshold and score the two maps
669 of Eq. (12) on held-out sessions. The two covariate maps are fixed
670 before the outcome sessions are inspected.
- 671 4. **Match physical controls.** Confirm that the observed coherence-to-
672 amplitude relation holds under the specific local diffusion D and ob-
673 servation noise. The model’s quadratic-variation rate $2D$ applies to
674 the latent phase; a band-limited analytic phase is smooth below its
675 correlation time, and on a constructed field its squared-increment rate
676 changes 37-fold between 0.2-ms and 20-ms steps. Estimating D there-
677 fore requires an explicit observation model for filtering and sampling.

678 Predeclare the recording window, regional fluctuation thresholds, and a
679 shared timing-error model. Fit the two scales and that observation model

on training sessions only; on held-out sessions, score the observed site-onset times and sites that never cross by the same regional predictive rule. Report site order as a secondary outcome and behavioural time-to-response separately. Sessions without a crossing are censored observations, not zero delays or exclusions. Compare the two scores only where independently estimated field and connectivity maps predict distinguishable onset patterns at the recording resolution. A held-out field-model advantage would support that specified spatial predictor; it would not identify an ephaptic cause, shared content or experience. Supplemental Material gives the controls and pilot quantities needed to judge whether the comparison is informative.

10 Discussion

10.1 Unity and a minimal self as interpretive motivation

We propose the state glued from compatible local descriptions as a formal correlate of conscious unity, and its accurate internal reconstruction as a correlate of a minimal self. In the cup example, this would mean one description that preserves the cup's relation to the perceiver as well as the scene. Compatibility permits distinct contents and uncertainty; the shared quantities must be chosen from independent evidence about what each region encodes. The reconstruction need not be a verbal or autobiographical account.

700 The minimal self, in Gallagher’s sense, is the pre-reflective sense of be-
 701 ing the subject of an experience, present before any narrative or reflective
 702 self-description (Gallagher, 2000). Two of its features motivate the proposed
 703 correlate. It is not a separate object of experience but a perspective built
 704 into the experienced scene; correspondingly, the reconstruction is not a sec-
 705 ond representation beside the glued state but a map whose fixed point is
 706 that state, read through a part of it. It is immediate rather than inferred
 707 from a higher-order state; correspondingly, the fixed point needs no further
 708 state that represents the first. The correspondence is structural. Self-model
 709 theory locates the phenomenal self in a transparent model a system has of
 710 itself (Metzinger, 2003), and interoceptive-inference accounts ground it in
 711 prediction of the body’s own states (Seth, 2013). The present condition is
 712 narrower than either: it asks only that a declared local encoding recover the
 713 global state, including perceiver-relative variables, at a declared fidelity, and
 714 it says nothing about transparency or about which bodily variables must be
 715 modelled. Either account could supply the family of states and the fidelity
 716 that the condition leaves as inputs. Under E89 that family is one state at
 717 zero error and a family of diameter at most $2\varepsilon/(1 - \Lambda)$ within error ε (Sec-
 718 tion 5.2), so under E89 the condition constrains a single self-state; a self that
 719 persists through change needs the input-conditioned map of Section 5.2. Its
 720 proposed relation to experienced perspective remains an interpretation. Nei-
 721 ther identification is among the connecting hypotheses of the composition
 722 (Table 1), and satisfying them would not measure experience. The unity

723 proposal is also only as definite as its cover. The gluing theorem holds for
724 whatever cover it is given, a disjoint cover satisfies it trivially, and the cover
725 of interest overlaps in content rather than in space. Section 4 proposes fixing
726 that cover by decodability on held-out data, which excludes covers chosen for
727 agreement and makes non-degeneracy a testable property of the recordings.
728 Its variables, criterion and decoder class are pre-registered by rules that never
729 consult agreement.

730 **10.2 A comparison that could disconfirm the link to** 731 **content**

732 The framework predicts that conscious content tracks compatibility, not co-
733 herence. The comparison must also separate compatibility from how well
734 each region encodes the stimulus, because two accurate decoders agree auto-
735 matically. If decoders d_A and d_B estimate a shared stimulus quantity x with
736 errors $e_A = d_A - x$ and $e_B = d_B - x$, then

$$|d_A - d_B| \leq |e_A| + |e_B|, \quad |d_A - d_B|^2 = |e_A|^2 + |e_B|^2 - 2\langle e_A, e_B \rangle. \quad (13)$$

737 Consciously perceived stimuli are usually decoded better, so higher agreement
738 on those trials would follow from single-region fidelity alone. At matched
739 single-region error, agreement differs only through the covariance of the two
740 errors. The comparison adds one premise to link compatibility to that term:
741 two regions describing one content err together, so that compatibility appears

742 as error covariance beyond what single-region fidelity implies.

743 Present matched stimuli that are perceived on some trials and not on
744 others, for example at masking or rivalry threshold, and fit decoders for one
745 shared stimulus quantity independently on two regions that both encode it,
746 selected by the decodability criterion of Section 4 on separate data. Clas-
747 sify trials by perceptual state with a no-report measure, such as reflexive
748 eye movements or pupil responses that track the dominant percept in rivalry
749 (Tsuchiya et al., 2015), so that the classification does not also select for the
750 neural correlates of reporting; report-based classification is a secondary anal-
751 ysis. Match the two classes on phase coherence between the regions and
752 on each region’s own decoding error (or, where the stimulus value is uncer-
753 tain, on each region’s signal strength). The statistic is the excess agreement
754 over a fidelity-matched null: pair region A ’s decode on one trial with region
755 B ’s decode on another trial of the same stimulus and error stratum, which
756 preserves both marginal accuracies and removes any coupling of the errors.

757 The proposed link predicts a larger excess on perceived trials at matched
758 coherence and matched single-region accuracy (Fig. 5). Equal excess across
759 classes, or excess that follows coherence and not perceptual state, counts
760 against it. A positive result is not yet specific: noise shared upstream of both
761 regions also correlates their errors, so the excess should be compared against
762 an upstream region’s decoding error, matched in the same way. A self-model
763 comparison has the same form: at matched accuracy for the scene variable,
764 the reconstruction of perceiver-relative variables, such as hand position rel-

765 active to the target, from a declared local encoding should be more accurate
 766 on perceived trials. These are correlational tests of proposed correlates, and
 767 the no-report classification is itself an indirect marker of experience.

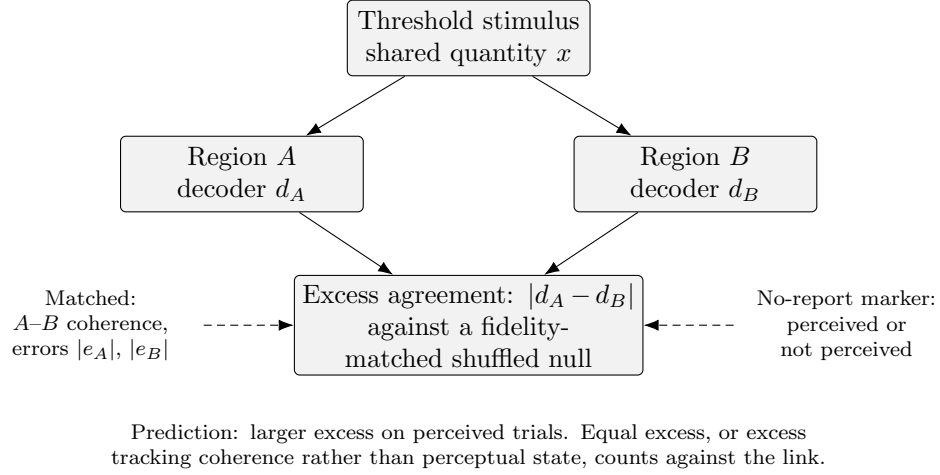


Figure 5: Design of the disconfirming comparison. Decoders for one shared stimulus quantity are fitted independently on two regions. Trials are classed by a no-report marker of perceptual state and matched on inter-regional phase coherence and on each region’s own decoding error, so that agreement cannot follow from single-region fidelity (Eq. (13)). The statistic is agreement in excess of a null that pairs decodes across trials within the same stimulus and error stratum.

768 10.3 What a test of the full proposal must establish

769 A test needs separate estimates of effective coupling and diffusion, shared
 770 content on overlapping regions, and reconstruction from a specified encoding
 771 at a declared fidelity. The cover, relevant state family and decoder must
 772 be fixed without selecting them for agreement. An independent experiential
 773 endpoint is needed to evaluate the interpretation; absence of a report alone

774 does not define absence of experience. A unified episode with incompati-
775 ble decoded descriptions would challenge the unity identification. A unified
776 episode lacking the proposed self-encoding would challenge the minimal-self
777 claim. Conversely, satisfying the mathematical conditions without such an
778 episode would challenge their sufficiency as correlates.

779 For the physical route, cortical actuator modes and prices, the geometry–
780 coupling relation and a dynamical onset model remain to be measured. The
781 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
782 diction. Controlled changes in coordination and independently decoded con-
783 tent agreement would address the representational connection. The synthetic
784 decoder exercise calibrates part of these measurements, and the EEG exercise
785 identifies the estimand a spatial test needs.

786 Integrated-information, global-workspace and predictive-processing ac-
787 counts organize unity, access and modelling differently (Tononi et al., 2016;
788 Baars, 1997; Dehaene et al., 2014; Friston, 2010). This account specifies con-
789 ditional relations among coupling, local content and reconstruction, each tied
790 to a measurement.

791 **11 Limitations**

792 *Experience and interpretation.* Synchrony is not itself an experiential mea-
793 sure, and nothing here establishes a consciousness criterion: the conditions
794 are proposed correlates to be tested against report or another independent

795 endpoint. The mathematical states are spatial measures and fields, and iden-
 796 tifying them with perceptual content requires independent decoding evidence.
 797 The constructions of Section 4 select neither the neural cover nor the decoder,
 798 and identifying their contents with perceptual content, or establishing that
 799 the identification bears on experience, is a further modelling task. The cor-
 800 respondence between the reconstruction's fixed point and the minimal self is
 801 structural and does not show that a fixed point is experienced as a perspec-
 802 tive. The remaining explanatory question is whether these relations occur in
 803 neural activity and matter for experience; the framework does not explain
 804 why satisfying them would be accompanied by experience.

805 *Formal scope.* The formal results are conditional on declared encoders,
 806 covers and maps whose cortical counterparts are unmeasured. Each edge
 807 of the composition is supplied rather than derived from the one above it,
 808 so the composition does not establish that self-representation follows from
 809 coherence. The joint witnesses satisfy every edge in a specified model: they
 810 prove neither that cortex satisfies those hypotheses nor that the glued state is
 811 experienced, and establish no common cortical mechanism. The contraction
 812 factor Λ is stipulated, not derived. It borrows the rate $(K - 2D)/2$ at which
 813 the first harmonic of the uniform density grows (Section 3), a rate about a
 814 different state from the one F acts on, and no result derives a Lipschitz law
 815 for an encoding or readout from the phase dynamics; constructing a map with
 816 that factor remains a modelling obligation. No physical input-conditioned
 817 map F_u is constructed, the gain L of Section 5.2 is derived here for no physical

818 readout, and E89 supplies no resolution of a declared family by the encoding
819 region. Agreement of a common projection does not establish agreement of
820 the full vector, and the general vector result of Section 4 supplies no identi-
821 fication with the spatial-measure sheaf. The cover on which compatibility is
822 assessed is fixed only relative to pre-registered variables, decoder class and
823 criterion, and the phase-to-content bounds are stated for spatial covers rather
824 than for the content cover that criterion yields. No result connects exact or
825 approximate compatibility to correlated decoding errors; the comparison of
826 Section 10.2 takes that link as a premise.

827 *Physical identification.* The formal field condition requires a continuous
828 kernel, a normalized substrate, a specified average coupling and noise, and a
829 site-contribution bound. Those conditions do not identify an electromagnetic
830 substrate; the extracellular interpretation needs the separate calibration of
831 Section 9, and the field commitment requires physical evidence beyond a
832 choice of mathematical representation. Phase organization under propofol
833 does not measure coupling, and the support comparisons in Supplemental
834 Material concern specified support constraints and supply neither that evi-
835 dence nor energy prices. Neither the conversion rate γ nor the installation
836 factor κ follows from a heat bound, and no cortical value of κ is known.
837 Testing the physical threshold requires independent estimates of the inter-
838 action kernel, its aggregation and diffusion in the same cortical regime. No
839 measured connectivity yet supplies the spectral route of Section 4.2. The
840 mean-field model has no spatially varying κ and derives no nucleation or-

841 der, which is why the onset comparison of Section 9.3 rests on a separately
842 specified model.

843 *Numerical and empirical controls.* The controls of Section 8 are finite
844 regimes on declared models and calibrate no cortical quantity. A slow con-
845 trol variable supplies no general guarantee of branch tracking; the winding
846 sweep establishes no transition and no biological identification; phase agree-
847 ment cannot replace an independently decoded content measurement. The
848 synthetic decoder exercise validates a measurement on known constructions,
849 and the EEG exercise checks a measurement pipeline: it identifies what a
850 pooled bipolar estimand cannot measure, and neither exercise identifies a
851 neural content code, tests experience or validates any experiential claim.

852 12 Conclusions

853 Phase coherence, compatibility and self-reconstruction are separable condi-
854 tions. Coherence constrains shared content only through encoders shared on
855 overlaps and with a bound that grows as the square root of the population
856 read. That bound is vacuous beyond 10 sites at the simulated sheet’s patch
857 order; chaining it through overlapping patches reaches 0.312 mm of that
858 sheet, and a spectral route removes the population count only for a declared
859 connectivity at a frequency-locked configuration. Across the centimetre sep-
860 arations of distinct cortical territories, the positive link from coherence to
861 content is the bridge assumption E78 rather than a derived result. Com-

862 patible local descriptions, not synchrony, determine one global state; and a
 863 contracting self-map fixes that state while reconstructing only a family of
 864 bounded diameter. Each result replaces an assumption about synchrony by
 865 a measurement: independently decoded agreement on overlaps, a declared
 866 encoding and its fidelity, and calibrated coupling and diffusion. The compar-
 867 ison of Section 10.2 states how such measurements could count against the
 868 proposed link to conscious content.

869 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	Eq. (1), Supp.	Phase diffusion in inverse time; in the Supplemental Material’s convergence proof, the phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring

Table 2 (continued)

Symbol	Defined	Meaning
U_{inst}, κ	Section 3.2	Installed energy of a declared actuator and the mode-data factor bounding its mean coupling by κU_{inst}
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (9)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (7)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time; in Eq. (4), τ is the duration of a change
J, σ_J, C	Eq. (4)	Probability current of the phase density, its current functional and a current budget
T	Section 5	Number of communication rounds to a deadline
ν	Supplemental Material	Fan-in of a communication graph: the most sites any one site hears from
$\gamma, \mathcal{T}, f$	Eq. (10)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	Section 9	Extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover's nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (9)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (7)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
E	Eq. (10)	Field amplitude, in the cortical identification
s, s_*	§4–5	A global section and the one the reconstruction fixes

870 **Data and code availability**

871 The Lean development, simulation scripts, tests and saved numerical sum-
872 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
873 `cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and Mathlib
874 commit `5ce203ecd53f8c3e47ee5530d7c86b94a55f0183`. Publication simu-
875 lation values are generated from saved summaries without rerunning pro-
876 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
877 `ds005620`, version 1.0.0 (Bajwa et al., 2024, 2025): subjects 1010, 1016, 1022,
878 1033, 1045, 1054, 1060, 1067, first resting block under sedation. The data,
879 subset and preprocessing are described in Supplemental Material and imple-
880 mented in `simulations/empirical_collapse.py`. The repository contains
881 the reproducible analysis and does not redistribute the raw recordings.

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885 **Use of AI tools**

886 AI assistance to this work was substantial and is not separable by task.
887 Several coding assistants and chat interfaces were used in an interleaved way
888 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,

889 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra and Sol; GPT-
 890 5.6 Terra, Luna and Sol), Claude Code (Claude Opus 5 and Opus 5.5),
 891 Gemini Pro through its web interface, and Anthropic’s Fable and DeepSeek
 892 v4 Flash through OpenRouter, accessed with the Hermes CLI. Between them
 893 they contributed to the Lean development, the simulation and analysis code,
 894 and the drafting and revision of both documents. Because the tools were
 895 used in combination and in alternation, no section, proof, figure or result is
 896 attributable to any one of them, and none is claimed to be. The author is
 897 responsible for all content, including everything a tool produced. Machine
 898 checking establishes the stated Lean results under their hypotheses; it does
 899 not validate biological interpretations or replace scientific review of generated
 900 code and prose.

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